

江苏大学全英文授课精品课程

Volumes

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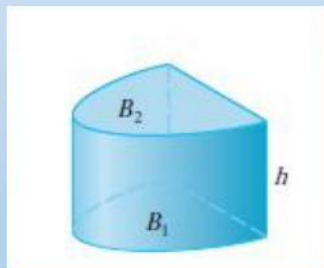
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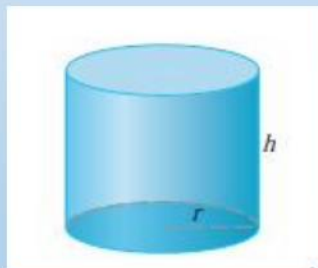
1 Volumes of Right Cylinder

Right Cylinder: a cylinder is bounded by a planeregion B_1 called the **base**, and a congruent region B_2 in a parallel plane. The cylinder consists of all points on line segents that are perpendicular to the base and join B_1 to B_2 .



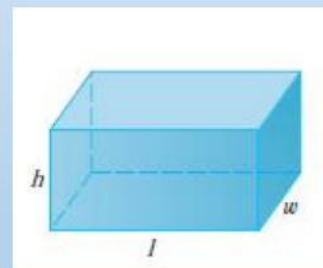
$$V = Ah$$

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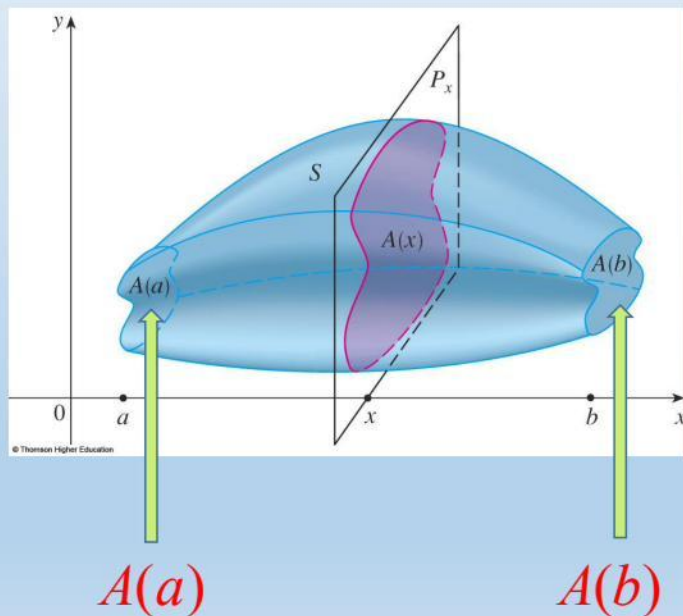
$$V = \pi r^2 h$$

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$$V = lwh$$

2 Volumes of Irregular Solids (with the Area of Cross-section Known)



Cross-section:

we intersect with a plane
and obtain a **plane region**,
the **cross-section** of solid S .

$A(x)$ will vary as x increases from a to b .



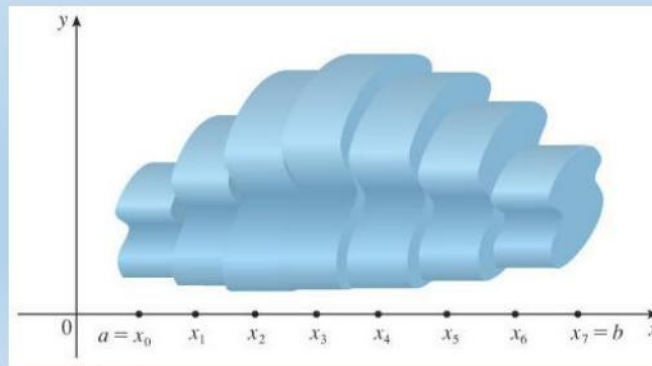
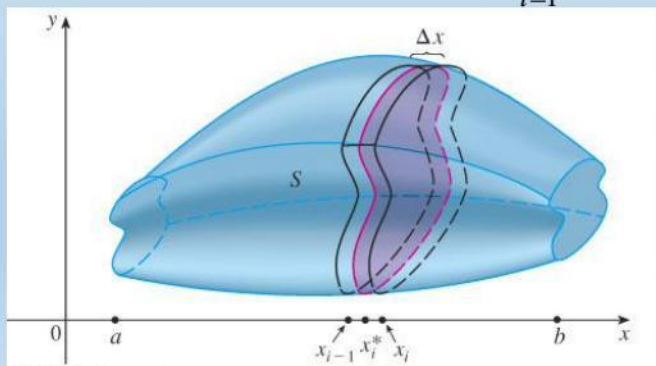
Divide : $a = x_0 < x_1 < \cdots < x_n = b$, $\Delta x = x_i - x_{i-1} = (b-a)/n$.

Divide S into n "slabs" by using the planes $P_{x_1}, P_{x_2}, \cdots, P_{x_n}$ to slice the solid.

$\Delta V_i \approx A(x_i^*) \Delta x$ an approximation to our intuitive conception of the volume of the i th slab

Take a sum: $V = \sum_{i=1}^n \Delta V_i \approx \sum_{i=1}^n A(x_i^*) \Delta x$ an approximation to the total volume

Take a limit: $V = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i^*) \Delta x = \int_a^b A(x) dx$





Definition of Volume:

Let S be a solid that lies between $x = a$ and $x = b$. If the cross-sectional area of S in the plane P_x , through x and perpendicular to the x -axis, is $A(x)$, where A is a continuous function, then the volume of S is:

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i^*) \Delta x = \int_a^b A(x) dx$$

How to get the $A(x)$? It is the area of a moving cross-section obtained by slicing through x perpendicular to the x -axis.

For the cylinder,
the cross-sectional area is constant $A(x) = A$ for all $x \in [a, b]$.

So our definition of the volume gives

$$V = \int_a^b A(x) dx = \int_a^b A dx = A(b - a)$$

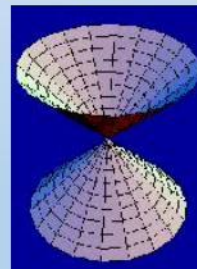
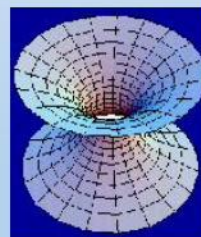
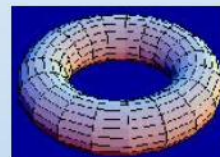
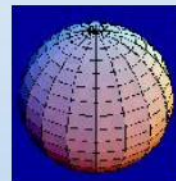
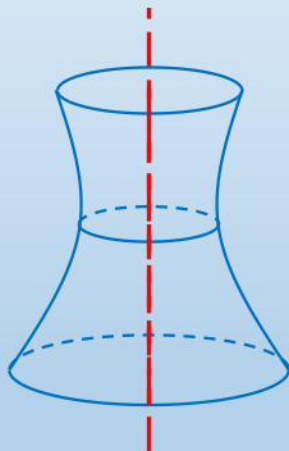


This agrees with the formula $V = Ah$



Volume of the Solids of Revolution

The **solids of revolution** are obtained by revolving a region about a line.



The cross - section obtained by slicing the solid perpendicular to the axis is
a circle (disk), or an annular(washer).

Example 1 Show that the volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.

Solution:

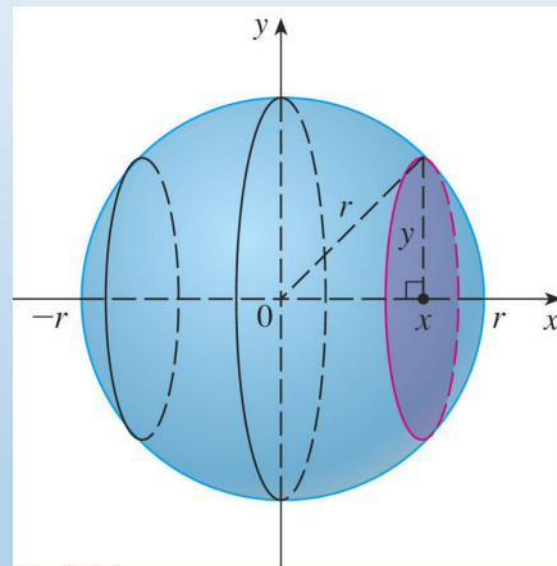
The radius of the cross-section is $r = y = \sqrt{r^2 - x^2}$.

The cross-section area is

$$A(x) = \pi y^2 = \pi(\sqrt{r^2 - x^2})^2 = \pi(r^2 - x^2)$$

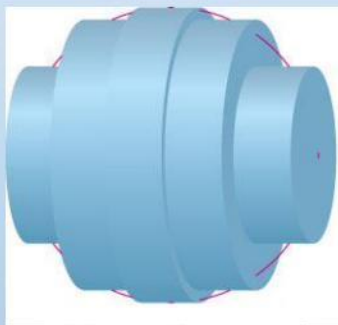
The volume of sphere is

$$\begin{aligned} V &= \int_{-r}^r A(x) dx = \int_{-r}^r \pi(r^2 - x^2) dx \\ &= 2 \int_0^r \pi(r^2 - x^2) dx = \frac{4}{3} \pi r^3 \end{aligned}$$

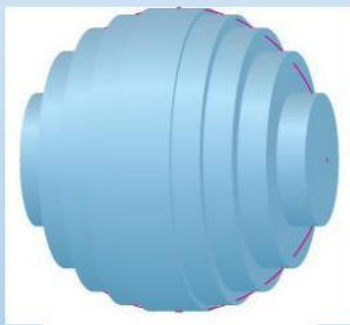


Example 1 Show that the volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.

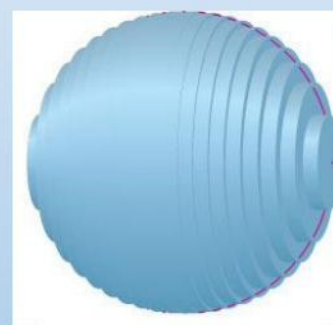
The three parts show the geometric interpretations of the Riemann sums when $n = 5, 10$ and 20 if we choose the sample points x_i^* to be the midpoints.



$n = 5, V \approx 4.2726$



$n = 10, V \approx 4.2097$



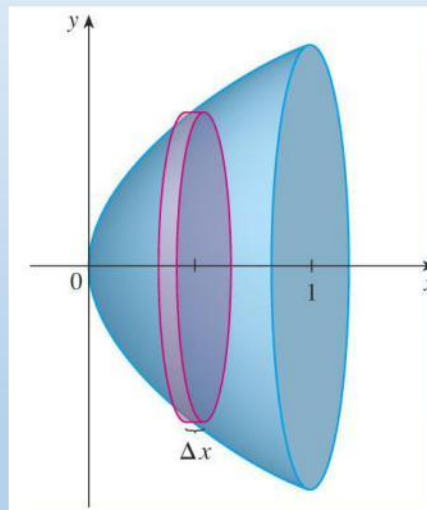
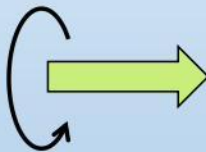
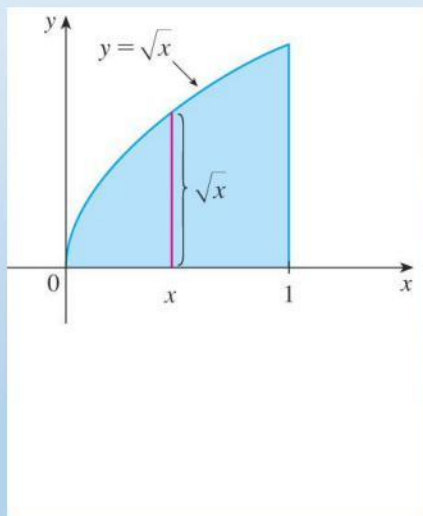
$n = 20, V \approx 4.1940$

Notice that as we increase the number of approximating cylinders, the corresponding Riemann sums become closer to the true volume.

Example 2

Find the volume of the solid obtained by rotating about the x -axis the region under the curve $y = \sqrt{x}$ from 0 to 1.

Solution:





Example 2

Find the volume of the solid obtained by rotating about the x -axis the region under the curve $y = \sqrt{x}$ from 0 to 1.

Solution:

When we slice through the point x , we get a disk with radius $r = \sqrt{x}$.

The area of the cross-section is:

$$A(x) = \pi(\sqrt{x})^2 = \pi x$$

The volume of the approximating cylinder is:

$$\Delta V = A(x)\Delta x = \pi x \Delta x$$

Δx is the thickness of this slice

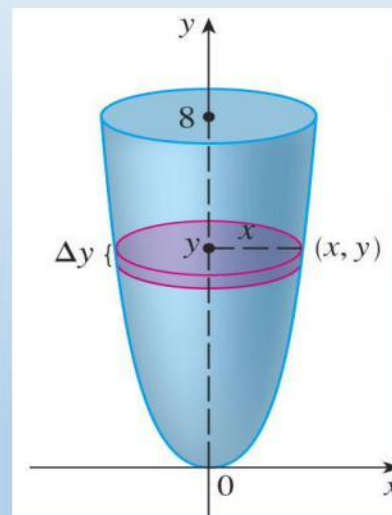
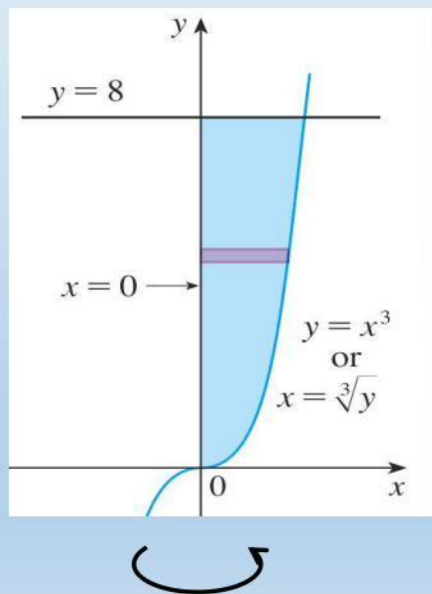
The solid lies between $x = 0$ and $x = 1$. so the volume is:

$$V = \int_0^1 A(x)dx = \int_0^1 \pi x dx = \left. \frac{\pi x^2}{2} \right|_0^1 = \frac{\pi}{2}$$

Example 3

Find the volume of the solid obtained by rotating the region bounded by $y = x^3$, $x = 0$ and $y = 8$ about y -axis.

Solution:





Example 3

Find the volume of the solid obtained by rotating the region bounded by $y = x^3$, $x = 0$ and $y = 8$ about y -axis.

Solution:

In this case, we slice through the point y , we get a disk with radius $r = \sqrt[3]{y}$.

The area of the cross-section is:

$$A(y) = \pi x^2 = \pi (\sqrt[3]{y})^2 = \pi y^{\frac{2}{3}}$$

The volume of the approximating cylinder is:

$$\Delta V = A(y) \Delta y = \pi y^{\frac{2}{3}} \Delta y$$

Δy is the thickness of this slice

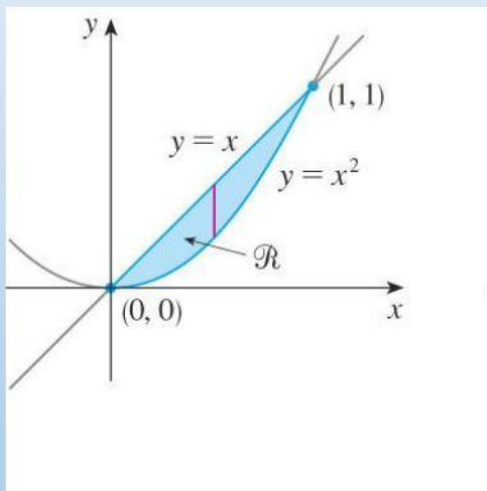
The solid lies between $y = 0$ and $y = 8$. so the volume is:

$$V = \int_0^8 A(y) dy = \int_0^8 \pi y^{\frac{2}{3}} dy = \frac{3\pi}{5} y^{\frac{5}{3}} \Big|_0^8 = \frac{96\pi}{5}$$

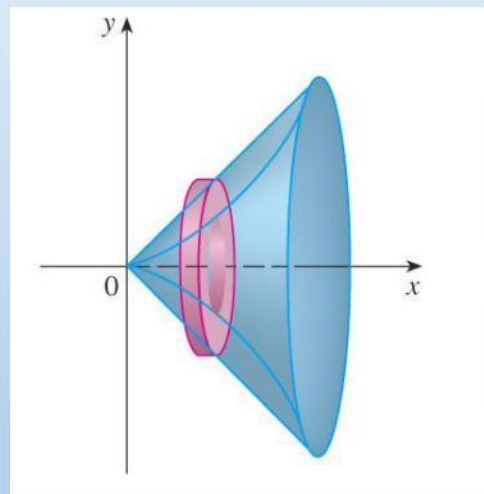
Example 4

Find the volume of the solid obtained by rotating the region bounded by $y = x^2$, $y = x$ about x -axis.

Solution:



The region between two curves

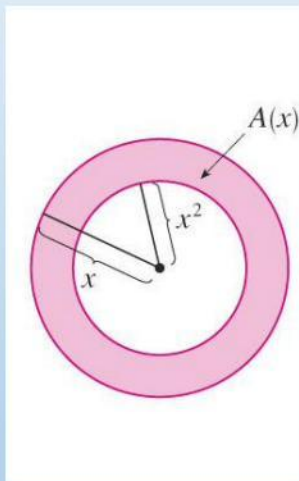


The solid of rotation



Example 4

Find the volume of the solid obtained by rotating the region bounded by $y = x^2$, $y = x$ about x -axis.



The cross-section perpendicular to the x -axis is an annular ring with inner radius $r_{in} = x^2$ and outer radius $r_{out} = x$.

The cross-section area is:

$$A(x) = \pi r_{out}^2 - \pi r_{in}^2 = \pi x^2 - \pi (x^2)^2$$

Therefore we have the volume:

$$\int_0^1 A(x) dx = \int_0^1 (\pi x^2 - \pi x^4) dx = \frac{2}{15} \pi$$

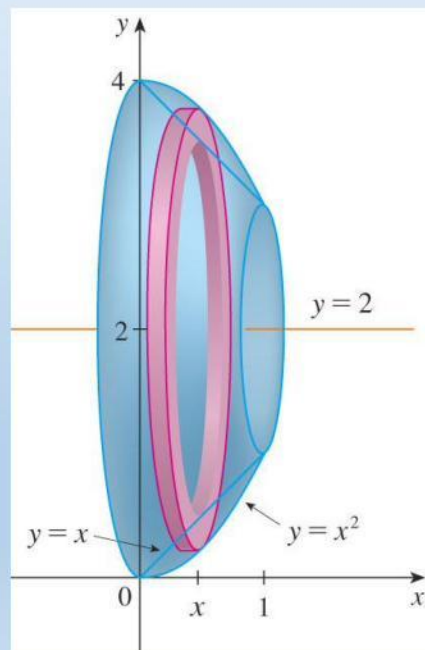
The cross-section



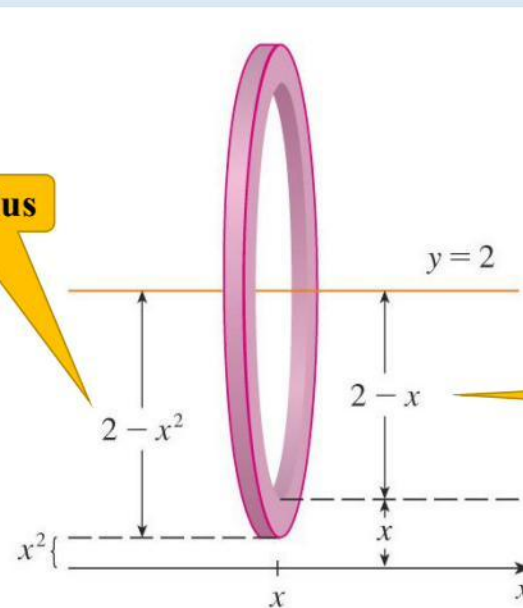
Example 5

Find the volume of the solid obtained by rotating the region bounded by $y = x^2$, $y = x$ about the line $y = 2$.

Solution:



outer radius



inner radius



Example 5

Find the volume of the solid obtained by rotating the region bounded by $y = x^2$, $y = x$ about the line $y = 2$.

The inner radius $r_{in} = 2 - x$ and outer radius $r_{out} = 2 - x^2$

The cross-section area is:

$$A(x) = \pi r_{out}^2 - \pi r_{in}^2 = \pi(2 - x^2)^2 - \pi(2 - x)^2$$

Therefore we have the volume:

$$\int_0^1 A(x) dx = \int_0^1 (\pi(2 - x^2)^2 - \pi(2 - x)^2) dx = \frac{8}{15} \pi$$





In general

The method of finding the area of revolution with the cross-section area $A(x)$ or $A(y)$:

***Case 1**

The cross-section is a **disk**,

we find the radius of disk (in terms of x or y) and the area

$$A = \pi(\text{radius})^2$$

***Case 2**

The cross-section is a **washer**(annular ring),

we find the inner radius and outer radius and the washer area

$$A = \pi(\text{outer radius})^2 - \pi(\text{inner radius})^2$$



Exercises 1 Find the volume of the solid obtained by revolving the region bounded by $y = x^2$, $y = 0$, $x = 1$ and $x = 2$ about:

- (a) x -axis (b) y -axis (c) $y = -1$ (d) $x = 3$





Exercises 2

Find the volume of the solid obtained by rotating the region bounded by $y = 2x$, $y = 0$ and $x = 1$ about:

- (a) x -axis (b) $x = 1$

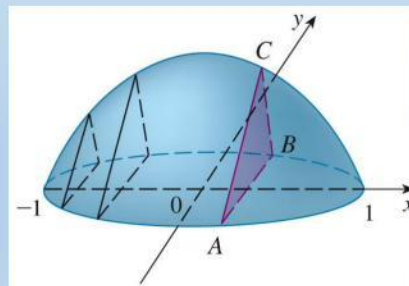
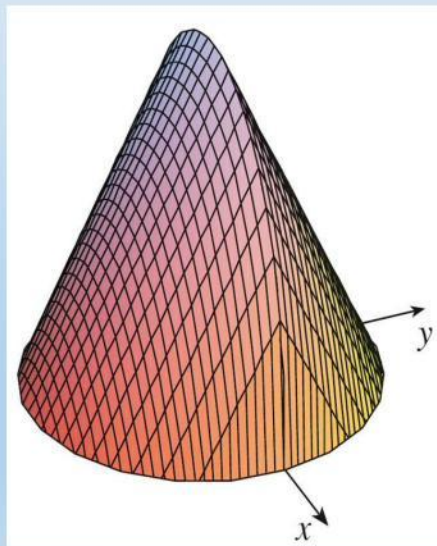


Volume of the Solids That are **NOT** Solids of Revolution

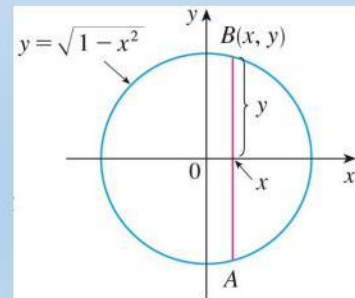
Example 6

The figure shows a solid with a circular base with radius $r = 1$. Parallel cross-sections perpendicular to the base are equilateral triangles. Find the volume of this solid.

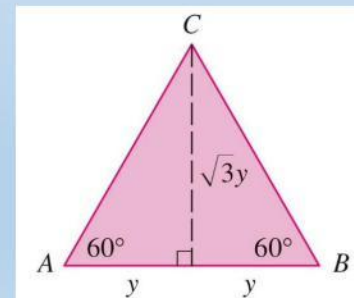
Solution: Let's take the circle to be $x^2 + y^2 = 1$



(a) The solid



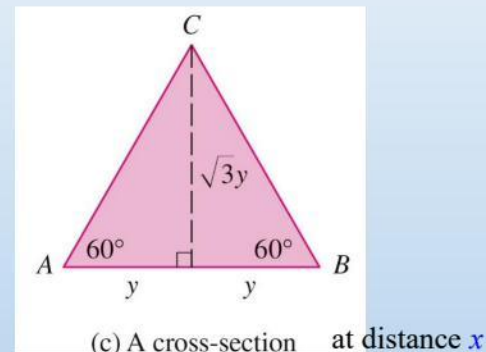
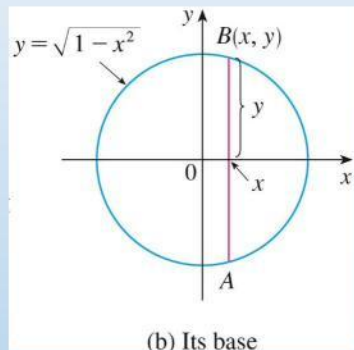
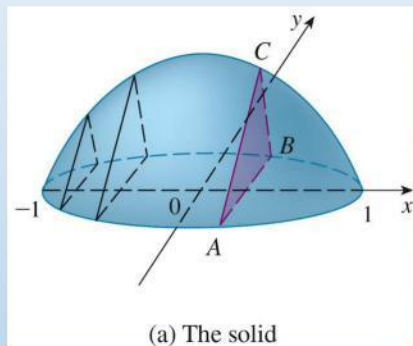
(b) Its base



(c) A cross-section
at distance x

Example 6

Let's take the circle to be $x^2 + y^2 = 1$



As B lies on the circle, we have $y = \sqrt{1 - x^2}$ So $|AB| = 2\sqrt{1 - x^2}$

height of the triangle ABC is $\sqrt{3}y = \sqrt{3}\sqrt{1 - x^2}$ (equilateral triangle)

Example 6

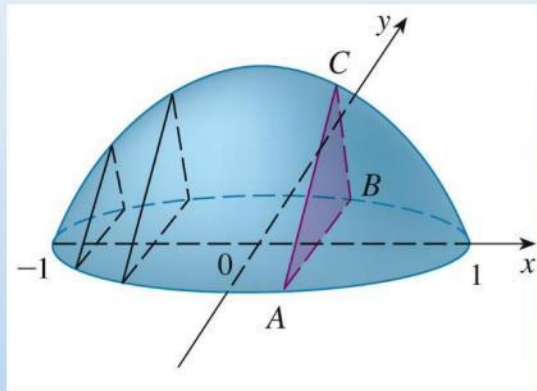
height is $\sqrt{3}y = \sqrt{3}\sqrt{1-x^2}$ (equilateral triangle)

Thus, the cross-sectional area is:

$$\begin{aligned} A(x) &= \frac{1}{2} \cdot 2\sqrt{1-x^2} \cdot \sqrt{3}\sqrt{1-x^2} \\ &= \sqrt{3}(1-x^2) \end{aligned}$$

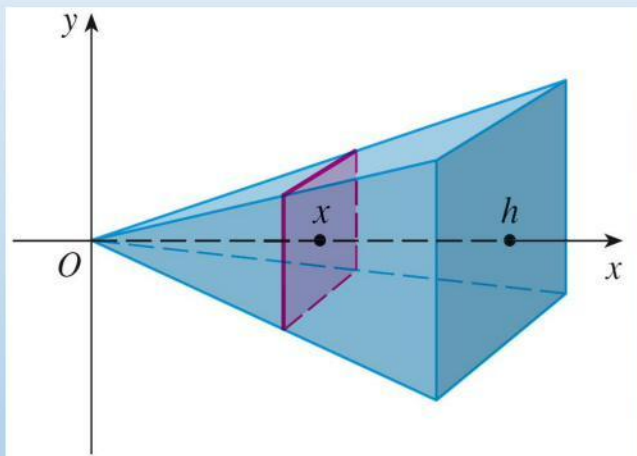
the volume of this solid is:

$$V = \int_{-1}^1 A(x) dx = \int_{-1}^1 \sqrt{3}(1-x^2) dx = 2 \int_0^1 \sqrt{3}(1-x^2) dx = \frac{4\sqrt{3}}{3}$$



Exercises 3

Find the volume of a pyramid whose base is a square with side L and whose height is h .



We place the origin O at the vertex of the pyramid and the x -axis along its central axis.

Any plane P_x that passes through x and is perpendicular to the x -axis intersects the pyramid in a square with side of length s .

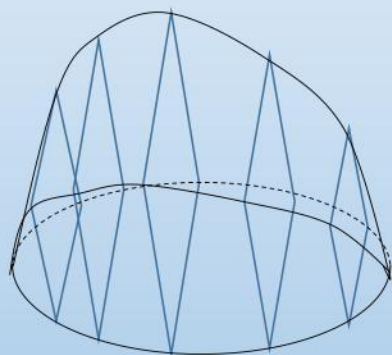
cross-section area $A(x) = ?$

volume of the pyramid $V(x) = \int_0^h A(x) dx$





Exercises 4 Find the volume of the solid whose base of S is a circle with radius r and parallel cross-sections perpendicular to the base are squares.





Exercises 5 A circular cylinder with the base radius r . A plane pass through the diameter of the base circle, and the angle between the plane and the base is α , please find the volume of the solid shown in the figuer.

